

Auteur: Ilyas Sefiani
Studierichting: Informatica
Oefeningenreeks 3C nr. 18.14

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{e^x \ln(1+3x)}{e^x - 1} &= \frac{0}{0} \\ &\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{e^x \ln(1+3x) + \frac{(1+3x)'}{1+3x} e^x}{e^x} \\ &= \lim_{x \rightarrow 0} \frac{e^x \ln(1+3x) + \frac{3}{1+3x} e^x}{e^x} \\ &= \frac{0+3}{1} \\ &= 3\end{aligned}$$

References